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ITA0609

Exp-11 : Write a program for the task of Credit Score Classification

```
# Import required libraries
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target
# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create the classifier
model = RandomForestClassifier(random_state=42)
# Train the model
model.fit(X_train, y_train)
# Predict the test data
y_pred = model.predict(X_test)
# Display results
print("Actual Labels:")
print(y_test)
print("\nPredicted Labels:")
print(y_pred)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
```

print("\nAccuracy:", accuracy * 100, "%")

Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)

Output:

```
Actual Labels:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Predicted Labels:
[1 0 0 1 1 0 0 0 0 1 1 0 1 0 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 1 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Accuracy: 96.49122807017544 %

Confusion Matrix:
[[40  3]
 [ 1 70]]
```